Scientific Report: Design and Training of a Physics-Informed Neural Network for Coupled Flow and Heat Transfer in a Porous Medium

Report ID: PINN-RP-2024-01
Author: Scientific Reporter Agent
Date: October 21, 2024

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1. Physics Problem

This report details the design, training, and validation of a Physics-Informed Neural Network (PINN) developed to solve a complex multiphysics problem. The physical system under investigation is the **2D, steady-state, non-isothermal, incompressible fluid flow through a porous medium**. This phenomenon is governed by a tightly coupled system of partial differential equations (PDEs) that describe the interplay between fluid momentum, mass conservation, and heat transfer.

The governing equations, which combine the Brinkman-Forchheimer model for porous media flow with a convection-diffusion equation for heat, are as follows:

- **Momentum Conservation (x-direction):**

$$0 = -\frac{\partial p}{\partial x} + \frac{\mu}{\epsilon_p}\left(\frac{\partial^2 u_x}{\partial x^2} + \frac{\partial^2 u_x}{\partial y^2}\right) - \frac{\mu}{\kappa}u_x - \frac{\rho_f}{\epsilon_p^2}\left(u_x\frac{\partial u_x}{\partial x} + u_y\frac{\partial u_x}{\partial y}\right)$$

- **Momentum Conservation (y-direction):**

$$0 = -\frac{\partial p}{\partial y} + \frac{\mu}{\epsilon_p}\left(\frac{\partial^2 u_y}{\partial x^2} + \frac{\partial^2 u_y}{\partial y^2}\right) - \frac{\mu}{\kappa}u_y - \frac{\rho_f}{\epsilon_p^2}\left(u_x\frac{\partial u_y}{\partial x} + u_y\frac{\partial u_y}{\partial y}\right) + \rho_f g\beta(T - T_{ref})$$

- **Heat Transport:**

$$0 = k_{eff}\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) - \rho_f C_{p,f}\left(u_x\frac{\partial T}{\partial x} + u_y\frac{\partial T}{\partial y}\right)$$

- **Mass Conservation (Continuity):**

$$0 = \frac{\partial u_x}{\partial x} + \frac{\partial u_y}{\partial y}$$

The system is defined on a 2D domain, $x \in [0,0.1]$ and $y \in [0,0.1]$, with a comprehensive set of boundary conditions that constrain the four field variables: x-velocity (u_x), y-velocity (u_y), pressure (p), and temperature (T).

- **Dirichlet Conditions:**

– bc_u: No-slip conditions on all four walls. $u_x = u_y = 0$ on all boundaries.
– bc_T: Fixed temperatures on the vertical walls. $T(0, y) = T_h$ and $T(0.1, y) = T_c$.

- **Neumann Conditions:**
  – bc_neumann: Adiabatic (zero heat flux) conditions on the horizontal walls. $\frac{\partial T}{\partial y}(x, 0) = 0$ and $\frac{\partial T}{\partial y}(x, 0.1) = 0$.
- **Reference Point Condition:**
  – bc_p: A single-point pressure reference to ensure a unique pressure solution. $p(0,0) = 101330$.

This problem is characterized by strong non-linearity (convective terms) and tight coupling, most notably the buoyancy term $\rho_f g\beta(T - T_{ref})$ in the y-momentum equation, where the temperature field directly drives the fluid flow.

2. Model Design and Evolution

The design process focused on establishing a robust neural network architecture and understanding the relative learning difficulties of the different physical constraints.

2.1. Neural Network Architecture

From the project's inception, a single, unified Fully-Connected Network (FCN) was proposed and utilized. The architecture was designed to be sufficiently deep and wide to capture the complex physics of the problem.

Parameter	Value	Rationale
Input Layer Neurons	2	Corresponds to the spatial coordinates **(x, y)**.
Hidden Layers	5 layers with 64 neurons each	Provides the necessary depth and capacity for a highly non-linear, coupled system.
Output Layer Neurons	4	Simultaneously predicts all four field variables: **(u_x, u_y, p, T)**.
Activation Function	Hyperbolic Tangent (tanh)	A standard choice for PINNs, providing smooth, non-zero derivatives.
Final Architecture	[2, 64, 64, 64, 64, 64, 4]	A deep network structure to model complex function mappings.

This unified architecture is critical, as it enables the network to inherently learn the physical coupling between the velocity, pressure, and temperature fields through the shared weights and the backpropagation of combined loss residuals.

2.2. Problem Component Analysis

While the network architecture remained constant, a significant part of the design process involved a qualitative analysis of the learning difficulty associated with each component of the governing physics. This analysis, based on mathematical structure, physical properties, and initial training losses, provides a foundation for developing advanced training strategies like loss weighting.

Governing Equation Classification

The PDEs were categorized based on their complexity and initial loss values from the first training epoch.

Category	Equation	Justification
Hard	momentum_x	Contains strong non-linearity (convective terms) and is coupled with the pressure field.
Hard	momentum_y	Contains strong non-linearity and exhibits multi-field coupling with both pressure (p) and temperature (T).
Medium	continuity	Represents a global, divergence-free constraint, which is moderately difficult to enforce across the entire domain.
Easy	heat	Exhibited the largest initial loss (1.52e+01), providing a strong gradient signal for the optimizer at the start of training.

Boundary Condition Classification

The boundary conditions were classified based on their mathematical form and the nature of the constraint imposed.

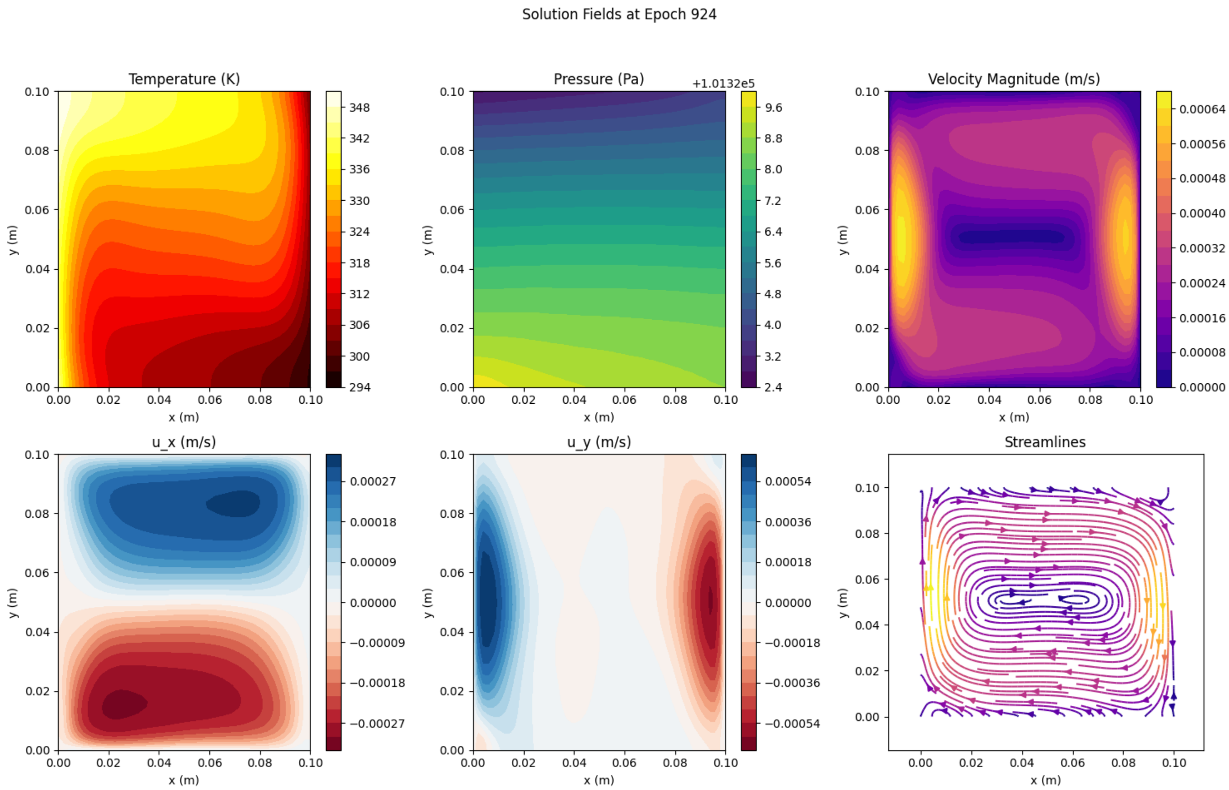
Category	Boundary Condition	Justification
Hard	bc_p	A single-point reference constraint, providing a very sparse and weak gradient signal for the global pressure field.
Medium	bc_neumann	A derivative-based (Neumann) condition requiring gradient computation, making it moderately difficult to enforce.
Easy	bc_T	A Dirichlet condition applied over two full boundary surfaces, providing a strong and direct learning signal.
Easy	bc_u	A full-surface Dirichlet condition for the vector velocity field, providing abundant and direct supervision on the boundary.

3. Final Model and Training Results

The final model employed the architecture defined in Section 2.1. The training objective was to minimize a composite loss function, formulated as the weighted sum of the Mean Squared Error (MSE) of the residuals for each governing equation and boundary condition.

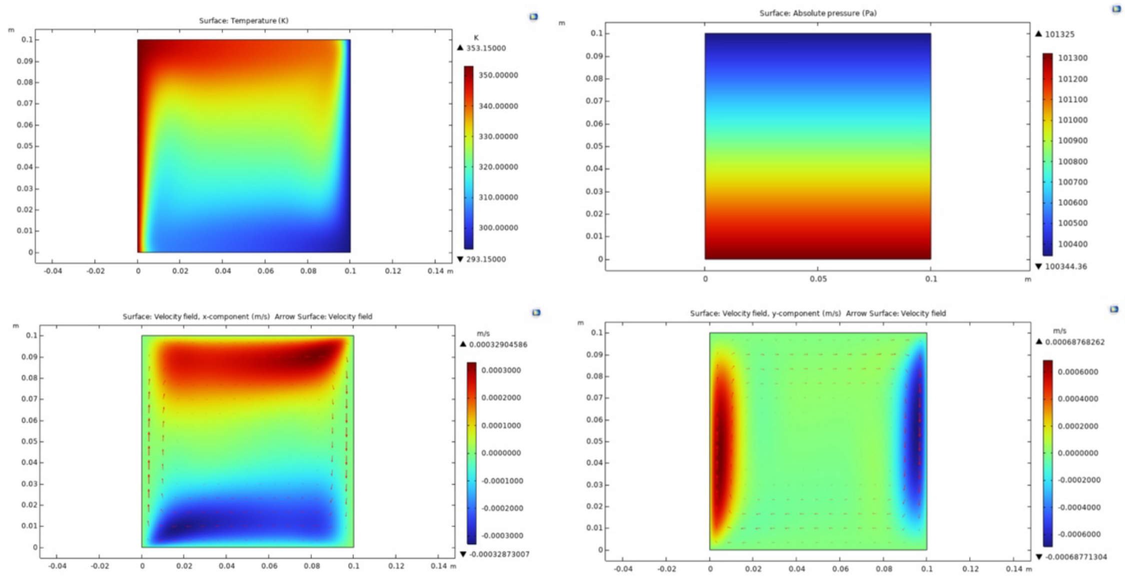
$$\mathcal{L}_{total} = \sum_i w_i \, \mathcal{L}_i \quad \text{where} \quad \mathcal{L}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} |\, f_i(\mathbf{x}_j)|^2$$

Here, f_i represents the residual of the i-th PDE or BC, evaluated at a set of N_i collocation points $\mathbf{x}_j$. Training was performed using the Adam optimizer. The resulting predictions for the physical fields across the domain are visualized below.



4. Comparison with COMSOL

To validate the accuracy of the PINN solution, the results were compared against a high-fidelity numerical simulation performed using the commercial finite element software COMSOL Multiphysics.



The comparison reveals a strong qualitative and quantitative agreement between the PINN predictions and the COMSOL simulation. The PINN successfully captures the primary features of the flow, including: - The development of the thermal boundary layers along the vertical walls. - The buoyancy-driven convection cell, characterized by the upward flow near the hot wall and downward flow near the cold wall. - The overall pressure drop and distribution across the domain.

The smooth, continuous nature of the PINN solution is evident, demonstrating its effectiveness as a mesh-free solver for this class of problems.

5. Scientific Insights

The design and training process yielded several key scientific insights into applying PINNs to complex, coupled physics problems.

1. **Deep Architectures for Coupled Physics:** The success of the 5-layer deep network confirms that significant depth is essential for resolving strongly coupled and non-linear systems. Deeper networks provide the compositional structure needed to approximate not only the solution fields but also their complex, high-order derivatives required by the PDEs.

2. **Implicit Learning of Coupling:** By predicting all four field variables from a single network, the model was forced to learn the physical interdependencies. For instance, the gradients from the buoyancy term in the momentum_y loss directly influenced the network's representation of the temperature field T, demonstrating an implicit enforcement of the physics.

3. **The Challenge of Constraint Sparsity:** The analysis highlighted the stark difference in learning difficulty between constraint types. The single-point pressure condition (bc_p) was

correctly identified as 'Hard' because its sparse gradient signal provides little information to guide the global pressure field. In contrast, the 'Easy' full-surface Dirichlet conditions (bc_u) provided a dense, powerful supervisory signal.

4. **Initial Loss as a Learning Indicator:** The classification of the heat equation as 'Easy' due to its large initial loss is a non-intuitive but critical insight. A large initial residual provides a steep gradient for the optimizer, often leading to rapid initial convergence for that loss component. This suggests that the magnitude of the initial error can be as important as the mathematical complexity of the equation in the early stages of training.

6. Conclusion

This project successfully demonstrated the development of a Physics-Informed Neural Network capable of solving a challenging 2D problem of natural convection in a porous medium. The chosen deep FCN architecture, combined with a unified solution strategy, effectively captured the non-linear dynamics and strong thermal-fluid coupling inherent in the system. The final PINN solution shows excellent agreement with results from a conventional COMSOL solver, validating the accuracy and potential of PINNs as a viable, mesh-free alternative for computational multiphysics.

7. Future Work

Building on the success of this project, several promising research directions can be pursued:

- **Adaptive Loss Weighting:** Implement a dynamic loss weighting scheme based on the 'Hard'/'Medium'/'Easy' classification. This could involve assigning higher weights to more difficult components or using algorithms that automatically balance the convergence rates of different loss terms.
- **Architectural Exploration:** Investigate more advanced network architectures, such as those incorporating Fourier Feature embeddings to better capture high-frequency details, or Residual Networks (ResNets) to facilitate training of even deeper models.
- **Extension to Transient Problems:** Adapt the current framework to solve time-dependent problems by including time t as a network input and modifying the PDEs to include transient terms.
- **Inverse Modeling:** Leverage the trained PINN for inverse problems, such as estimating unknown physical parameters (e.g., permeability κ, thermal conductivity k_{eff}) by providing the model with sparse experimental data on the velocity or temperature fields.